In [1]:
```
import numpy as np
import pandas as pd
df=pd.read_csv(r"C:\Users\kyoga\Downloads\Social_Network_Ads - Social_Net
df
```

Out[1]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>User ID</th> <th>Gender</th> <th>Age</th> <th>EstimatedSalary</th> <th>Purchased</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> <th>...</th> <th>395</th> <th>396</th> <th>397</th> <th>398</th> <th>399</th> </tbody> <tbody></tbody>

15624510	Male	19	19000	0
15810944	Male	35	20000	0
15668575	Female	26	43000	0
15603246	Female	27	57000	0
15804002	Male	19	76000	0
...	...	...	...	...
15691863	Female	46	41000	1
15706071	Male	51	23000	1
15654296	Female	50	20000	1
15755018	Male	36	33000	0
15594041	Female	49	36000	1

400 rows × 5 columns

In [2]:
```
df.head()
```

Out[2]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>User ID</th> <th>Gender</th> <th>Age</th> <th>EstimatedSalary</th> <th>Purchased</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> </tbody> <tbody></tbody>

15624510	Male	19	19000	0
15810944	Male	35	20000	0
15668575	Female	26	43000	0
15603246	Female	27	57000	0
15804002	Male	19	76000	0

In [3]:
```
features=df.iloc[:,[2,3]].values
label=df.iloc[:,4].values
features
```

Out[3]:
```
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```

In [4]:
```
label
```

Out[4]:
```
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       0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0,
       0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0,
       0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0,
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       0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0,
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       1, 1, 0, 1])
```

In [10]:
```
%pip install scikit-learn
```

```
Requirement already satisfied: scikit-learn in .\AppData\Local\Python\pyth
oncore-3.14-64\Lib\site-packages (1.9.1)
Requirement already satisfied: numpy>=1.24.1 in .\AppData\Local\Python\pyt
honcore-3.14-64\Lib\site-packages (from scikit-learn) (2.5.2)
Requirement already satisfied: scipy>=1.10.0 in .\AppData\Local\Python\pyt
honcore-3.14-64\Lib\site-packages (from scikit-learn) (1.18.1)
Requirement already satisfied: joblib>=1.4.0 in .\AppData\Local\Python\pyt
honcore-3.14-64\Lib\site-packages (from scikit-learn) (1.6.0)
Requirement already satisfied: narwhals>=2.0.1 in .\AppData\Local\Python\p
ythoncore-3.14-64\Lib\site-packages (from scikit-learn) (2.27.1)
Requirement already satisfied: threadpoolctl>=3.5.0 in .\AppData\Local\Pyt
hon\pythoncore-3.14-64\Lib\site-packages (from scikit-learn) (3.7.0)
Requirement already satisfied: cloudpickle>=3.0 in .\AppData\Local\Python
\pythoncore-3.14-64\Lib\site-packages (from joblib>=1.4.0->scikit-learn)
(3.1.2)
Note: you may need to restart the kernel to use updated packages.
```

In [13]:
```
%whos
```

```
Variable                Type        Data/Info
------------------------------------------
LogisticRegression      type        <class 'sklearn.linear_mo<...>stic.Log
isticRegression'>
classification_report   function    <function classification_<...>rt at 0x
0000022FC79BEF00>
i                       int         1
train_test_split        function    <function train_test_split at 0x000002
2FC7A27C10>
```

In [6]:
```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression

print("Import successful")
```

```
Import successful
```

In [12]:
```
import pandas as pd

df = pd.read_csv(r"C:\Users\kyoga\Downloads\logistic_regression_features_
```

```
print(df.head())
print(df.columns)
```

```
   Hours_Studied  Attendance_Percent  Previous_Score  Pass_Label
0              2                  55              35           0
1              3                  60              40           0
2              1                  50              30           0
3              4                  65              45           0
4              2                  70              38           0
Index(['Hours_Studied', 'Attendance_Percent', 'Previous_Score', 'Pass_Labe
l'], dtype='str')
```

In [ ]:
```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import classification_report
```